

Chapter 4.

Sequences – some specific types.

Suppose a business buys an item of machinery costing \$250 000. Whilst this purchase will mean that the money the business has in the bank will decrease by \$250 000, the business now has an asset worth \$250 000. Hence the business has not lost \$250 000, it has just re-allocated it from money in the bank to machinery. However, one year later the machine will no longer be new. Its value is likely to have **depreciated**. The company no longer has an asset worth \$250 000. For accounting purposes the business will need to be able to quote a value for the machine but how much will it be worth after one year, two years, three years etc? Consider each of the following situations:

Situation One. Depreciation as a fixed annual amount.

(Flat rate, or straight line, method of depreciation.)

Let us suppose that the machine, initially worth \$250 000, depreciates by the same amount each year and that after five years it is only worth its "scrap value" of \$25 000.

Copy and complete:	Initial value:	\$250 000
	Value after 1 year	_____
	Value after 2 years	_____
	Value after 3 years	_____
	Value after 4 years	_____
	Value after 5 years	\$25 000

Situation Two. Depreciation by the number of units produced.

(Unit cost, or units of production, method of depreciation.)

Suppose instead that the machine is expected to produce 600 000 units, after which it will only have scrap value of \$25 000, depreciation occurring at a fixed rate per unit produced.

Copy and complete:	Initial value:	\$250 000
	Value after 100 000 units produced	_____
	Value after 200 000 units produced	_____
	Value after 300 000 units produced	_____
	Value after 400 000 units produced	_____
	Value after 500 000 units produced	_____
	Value after 600 000 units produced	\$25 000

Situation Three. Depreciation by a percentage of the declining value.

(Declining balance, or reducing balance, method of depreciation.)

Suppose now that the value of the machine at the end of each year is 80% of its value at the beginning of that year, i.e. a depreciation rate of 20%.

Copy and complete:	Initial value:	\$250 000
	Value after 1 year	_____
	Value after 2 years	_____
	Value after 3 years	_____

Situation one and situation two on the previous page each involved the amount reducing at a fixed amount, either per year or per 100 000 units.

Situation three involved the amount reducing by virtue of repeated multiplication by a number, in this case 0.8. This number being between 0 and 1 producing the sequence of amounts that were decreasing.

Sequences of numbers for which each term, after the first, is obtained by adding a constant amount to the previous term are called **arithmetic sequences**.

Sequences of numbers for which each term, after the first, is obtained by multiplying the previous term by a constant amount are called **geometric sequences**.

Arithmetic sequences.

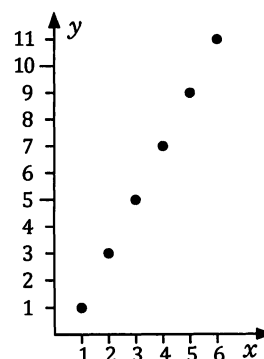
Notice that in the table of values on the right, as the x -values increase by 1 the y -values increase by 2. As we would expect from this constant first difference pattern of 2, graphing gives points that lie in a **straight line** and the gradient of the line is 2.

As mentioned above, sequences like 1, 3, 5, 7, 9, ... in which each term is obtained from the previous term by the addition of some constant number are said to be

Arithmetic Sequences.

These are also known as *Arithmetic Progressions* or simply *APs*.

x	1	2	3	4	5	6
y	1	3	5	7	9	11



For example, the AP: 1, 2, 3, 4, 5, 6, ...

In this case we say that **the first term** is 1 and **the common difference** is 1.

Similarly for the AP: 1, 3, 5, 7, 9, 11, ...
we say that the first term is 1 and the common difference is 2.

For 7, 11, 15, 19, 23, 27, ...
the first term is 7 and the common difference is 4.

For 67, 62, 57, 52, 47, 42, ...
the first term is 67 and the common difference is -5.

Thus all arithmetic sequences are of the form:

$$a, \quad a + d, \quad a + 2d, \quad a + 3d, \quad a + 4d, \quad a + 5d, \quad a + 6d, \quad \dots$$

In this general form we have a first term of "a" and common difference "d".

Hence, to define this general arithmetic sequence recursively:

$$T_{n+1} = T_n + d, \quad T_1 = a.$$

(Sometimes written with lower case t:

$$t_{n+1} = t_n + d \quad t_1 = a.)$$

An arithmetic sequence is such that $T_{n+1} = T_n + 5$ and the first term, T_1 , is 7. Find the first four terms of the sequence.

Hence if $T_1 = 7$ it follows that $T_2 = 7 + 5$

$$= 12$$

$$T_3 = 12 + 5$$

= 17

$$T_4 = 17 + 5$$

= 22

For each of the following sequences state whether the sequence is an AP or not and, for those that are, state the first term, the common difference and a recursive formula.

- | | | | | | | | |
|-----|-----|-----|-----|-----|-----|-----|-----|
| (a) | 13, | 18, | 23, | 28, | 33, | 38, | ... |
| (b) | 3, | 6, | 12, | 24, | 48, | 96, | ... |
| (c) | 90, | 79, | 68, | 57, | 46, | 35, | ... |

- Thus the sequence is an Arithmetic Progression with first term = 13

the common difference = 5

and $T_{n+1} = T_n + 5$

- Thus the sequence is not an Arithmetic Progression.

- Thus the sequence is an Arithmetic Progression with first term = 90

the common difference = -11

and $T_{n+1} = T_n - 11$

Example 3

A sequence is defined by $T_n = 2T_{n-1} + 1$ with $T_1 = 3$. Determine the first five terms of the sequence and hence determine whether the sequence is an arithmetic sequence.

The formula $T_n = 2T_{n-1} + 1$ tells us that each term is obtained by multiplying the previous term by 2 and then adding 1.

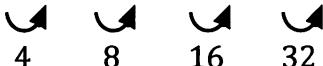
Thus if $T_1 = 3$ it follows that

$$\begin{aligned} T_2 &= 2(3) + 1 = 7, \\ T_3 &= 2(7) + 1 = 15, \\ T_4 &= 2(15) + 1 = 31, \\ T_5 &= 2(31) + 1 = 63. \end{aligned}$$

The first five terms are 3, 7, 15, 31, 63.

These terms do not have a common difference.

The sequence is not an arithmetic sequence.

3 7 15 31 63

 4 8 16 32

Alternatively, and as mentioned in the previous chapter, a calculator or computer spreadsheet can be used to display the terms of a sequence (and the partial sums if so desired) once an appropriate recursive rule and first term have been entered.

☒ $a_{n+1} = 2 \cdot a_n + 1$ $a_1 = 3$		
☐ $b_{n+1} = \square$ $b_1 = 0$		
☐ $c_{n+1} = \square$ $c_1 = 0$		
n	a_n	Σa_n
1	3	3
2	7	10
3	15	25
4	31	56
5	63	119

	A	B	C	D
1	3	3		
2	7	10		
3	15	25		
4	31	56		
5	63	119		
6	127	246		
7	255	501		
8	511	1012		
9	1023	2035		
10	2047	4082		

$=2 \cdot A3 + 1$

$=\text{Sum}(A\$1:A6)$

What does the inclusion of the \$ symbol do?

Create the spreadsheet yourself and use the "fill down" ability when creating it.

Geometric sequences.

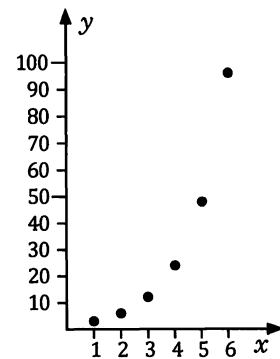
Notice that in the table of values on the right, as the x -values increase by 1 the y -values multiply by 2. Graphing the values gives the shape shown on the right which you may recognise as the characteristic shape of an **exponential** graph.

As mentioned previously, sequences that progress by each term being obtained by multiplying the previous term by a constant number are said to be

Geometric Sequences.

These are also known as Geometric Progressions or simply GPs.

x	1	2	3	4	5	6
y	3	6	12	24	48	96



For example the GP 5, 15, 45, 135, 405, 1215, ...
In this case we say that **the first term** is 5 and **the common ratio** is 3.

Similarly for the GP: 0.5, 1, 2, 4, 8, 16, ...
we say that the first term is 0.5 and the common ratio is 2.

For 1000, 100, 10, 1, 0.1, 0.01, ...
the first term is 1000 and the common ratio is 0.1.

For 64, 96, 144, 216, 324, 486, ...
the first term is 64 and the common ratio is 1.5.

Thus all GPs are of the form:

$a, ar, ar^2, ar^3, ar^4, ar^5, ar^6, \dots$

In this general form we have a first term of "a" and common ratio "r".

Using recursive notation we have $T_{n+1} = r \times T_n$ with $T_1 = a$, (or $t_{n+1} = r t_n$, $t_1 = a$).

Example 4

For each of the following sequences state whether the sequence is a geometric sequence or not and, for those that are, state the first term, the common ratio and a recursive formula.

- (a) 3, 6, 12, 24, 48, 96, ... ,
 (b) 128, 96, 72, 54, 40.5, 30.375, ... ,
 (c) 4, 9, 14, 19, 24, 29, ... ,

(a) $\frac{6}{3} = 2$, $\frac{12}{6} = 2$, $\frac{24}{12} = 2$, $\frac{48}{24} = 2$, $\frac{96}{48} = 2$.

Each term is the previous term multiplied by 2.

Thus the sequence is a geometric sequence with first term = 3

the common ratio = 2

and $T_{n+1} = 2T_n$

(or $T_n = 2T_{n-1}$)

- (b) The given sequence is: 128, 96, 72, 54, 40.5, 30.375, ... ,
 $\frac{96}{128} = 0.75$, $\frac{72}{96} = 0.75$, $\frac{54}{72} = 0.75$, $\frac{40.5}{54} = 0.75$, $\frac{30.375}{40.5} = 0.75$.

Each term is the previous term multiplied by 0.75.

Thus the sequence is a geometric sequence with first term = 128

the common ratio = 0.75

and $T_n = 0.75T_{n-1}$

- (c) The given sequence is 4, 9, 14, 19, 24, 29, ... ,
 $\frac{9}{4} = 2.25$, $\frac{14}{9} = 1.\bar{5}$.

The sequence does not have a common ratio.

Thus the sequence is not a geometric sequence.

Example 5

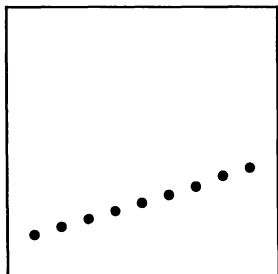
\$400 is invested in an account and earns \$20 interest each year.

- (a) How much is the account worth after 1 year, 2 years, 3 years and 4 years?
 (b) Do the amounts the account is worth at the end of each year form an arithmetic sequence, a geometric sequence or neither of these?

- | | | | | | | |
|-----|---------------|--------------------|---------------------|---------------------|---------------------|-----|
| (a) | Initial value | value after 1 year | value after 2 years | value after 3 years | value after 4 years | ... |
| | \$400 | \$400 + 1(\$20) | \$400 + 2(\$20) | \$400 + 3(\$20) | \$400 + 4(\$20) | ... |
| | \$400 | \$420 | \$440 | \$460 | \$480 | ... |
- After 1, 2, 3 and 4 years the account is worth \$420, \$440, \$460 and \$480 respectively.

- (b) The situation gives rise to amounts with a common difference of \$20.
 The amounts the account is worth at the end of each year form an arithmetic sequence.

As you are aware, the situation described in the previous example is not the way that an investment usually earns interest. Once the \$20 interest has been added at the end of year 1 the account has \$420 in it and it is this \$420 that attracts interest in year 2, not just the initial \$400. In this way the interest earned in one year itself attracts interest in subsequent years, i.e. compound interest is involved, rather than the simple interest situation described in the previous example.



Example 6

\$2000 is invested and accrues interest at a rate of 10% per annum, compounded annually.

- (a) If no further deposits are made how much will be in the account after 1 year, 2 years, 3 years and 4 years?
- (b) Do the amounts the account is worth at the end of each year form an arithmetic sequence, a geometric sequence or neither of these?

(a)	Initial value	value after 1 year	value after 2 years	value after 3 years	value after 4 years	...
	\$2000	$\$2000 \times 1.1$	$\$2000 \times 1.1^2$	$\$2000 \times 1.1^3$	$\$2000 \times 1.1^4$	...
	\$2000	\$2200	\$2420	\$2662	\$2928.20	...

After 1, 2, 3 and 4 years the account is worth \$2200, \$2420, \$2662 and \$2928.20 respectively.

- (b) The situation gives rise to amounts with a common ratio of 1.1.
The amounts the account is worth at the end of each year form a geometric sequence.

Exercise 4A

For each of the following arithmetic sequences state

- the first term, T_1 ,
- and • the $(n + 1)^{\text{th}}$ term, T_{n+1} , in terms of the n^{th} term, T_n .

1. 6, 10, 14, 18, 22, 26, ...
2. 28, 26, 24, 22, 20, 18, ...
3. 5, 15, 25, 35, 45, 55, ...
4. 7.5, 10, 12.5, 15, 17.5, 20, ...
5. 100, 89, 78, 67, 56, 45, ...

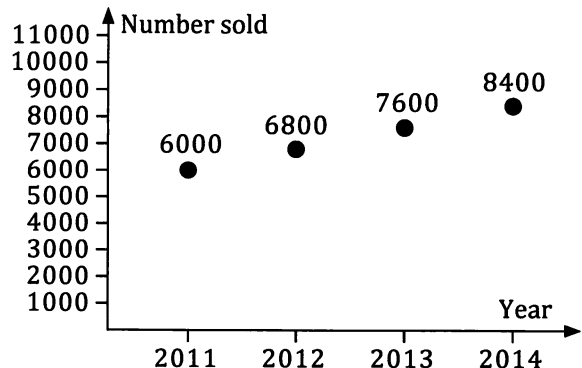
For each of the following geometric sequences state

- the first term, T_1 ,
- and • the n^{th} term, T_n , in terms of the $(n - 1)^{\text{th}}$ term, T_{n-1} .

6. 6, 12, 24, 48, 96, 192, ...
7. 0.375, 1.5, 6, 24, 96, 384, ...
8. 384, 96, 24, 6, 1.5, 0.375, ...
9. 50, 150, 450, 1350, 4050, 12150, ...
10. 1000, 1100, 1210, 1331, 1464.1, 1610.51, ...

For each of the following sequences state whether the sequence is arithmetic, geometric or neither of these two types.

11. 2, 5, 11, 23, 47, 95, ...
12. 1, 5, 25, 125, 625, 3125, ...
13. 13, 14.5, 16, 17.5, 19, 20.5, ...
14. 50, 39, 28, 17, 6, -5, ...
15. 1, 1, 2, 3, 5, 8, ...
16. 128, 160, 200, 250, 312.5, 390.625, ...
17. $T_{n+1} = 3T_n$, $T_1 = 3$.
18. $T_{n+1} = T_n + 6$, $T_1 = 2$.
19. $T_{n+1} = 3T_n + 5$, $T_1 = 1$.
20. $T_n = (T_{n-1})^2$, $T_1 = 7$.
21. $T_n = T_{n-1} - 8$, $T_1 = 2\ 000$.
22. $T_n = (0.5)T_{n-1}$, $T_1 = 8$.
23. An AP has a first term of 8 and a common difference of 3. Determine the first four terms of the sequence and the recursive rule for T_{n+1} in terms of T_n .
24. An AP has a first term of 100 and a common difference of -3. Determine the first four terms of the sequence and the recursive rule for T_{n+1} in terms of T_n .
25. A GP has a first term of 11 and a common ratio of 2. Determine the first four terms of the sequence and the recursive rule for T_{n+1} in terms of T_n .
26. A GP has a first term of 2048 and a common ratio of 0.5. Determine the first four terms of the sequence and the recursive rule for T_{n+1} in terms of T_n .
27. The graph on the right shows the number of vehicles a company sold in a particular country each year from 2011 to 2014.
 - (a) Verify that the figures for these years are in arithmetic progression.
 - (b) With $N_{2011} = 6000$ write a recursive rule for N_{n+1} in terms of N_n .



28. Each term of a sequence is obtained using the recursive rule

$$T_n = T_{n-1} + 10\% \text{ of } T_{n-1}$$

- (a) Is the sequence an arithmetic progression, a geometric progression or neither of these?
 (b) If the first term of the sequence is 500 find the next three terms.

29. Each term of a sequence is obtained using the recursive rule

$$T_n = T_{n-1} + 25\% \text{ of } T_{n-1}$$

- (a) Is the sequence an arithmetic progression, a geometric progression or neither of these?
 (b) If the first term of the sequence is 1000 find the next three terms.

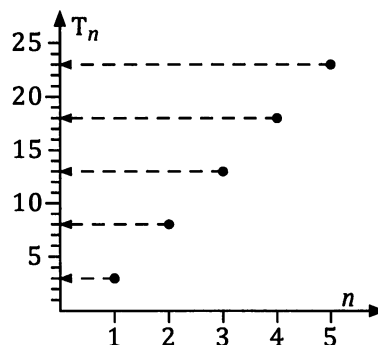
30. Each term of a sequence is obtained using the recursive rule

$$T_n = T_{n-1} - 10\% \text{ of } T_{n-1}$$

- (a) Is the sequence an arithmetic progression, a geometric progression or neither of these?
 (b) If the first term of the sequence is 24000 find the next three terms.

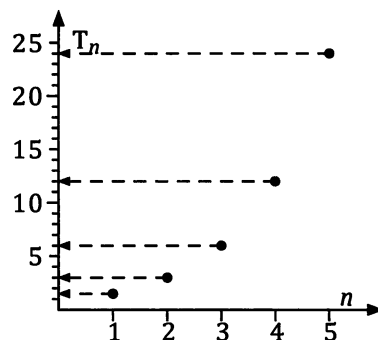
31. The graph on the right indicates the first five terms, T_1 to T_5 , of a sequence, all of which are whole numbers.

- State (a) the first term and a recursive rule for the sequence,
 (b) whether the sequence is arithmetic, geometric or neither of these types.



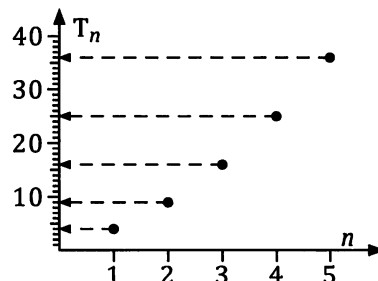
32. The graph on the right indicates the first five terms, T_1 to T_5 , of a sequence. The last four of these terms are whole numbers.

- State (a) the first term and a recursive rule for the sequence,
 (b) whether the sequence is arithmetic, geometric or neither of these types.

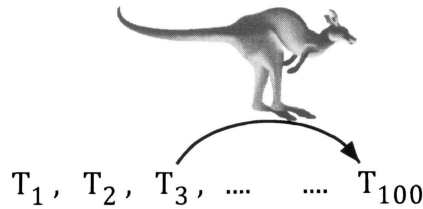


33. The graph on the right indicates the first five terms, T_1 to T_5 , of a sequence. All of these terms are whole numbers.

- Determine (a) the first five terms,
 and (b) whether the sequence is arithmetic, geometric or neither of these types.



34. \$1200 is invested in an account and earns \$96 interest each year.
- (a) How much is the account worth after 1 year, 2 years, 3 years and 4 years?
 - (b) Are these amounts in arithmetic progression, geometric progression or neither of these?
 - (c) Express the sequence of values:
Initial value, value after 1 year, value after 2 years, ...
using recursive notation.
35. Won Yim starts working for a particular company on the 1st January one year and is paid an initial annual salary of \$45 000 with a guaranteed \$1 500 rise each year for the next 7 years.
- Express the sequence of annual salaries over this time as a sequence using recursive notation and state whether the terms of the sequence progress arithmetically, geometrically or neither of these.
36. Joe started a new job on 1st January 2014 and, during 2014, he received a salary of \$68 000. His contract guarantees a salary increase of 5% of the salary of the previous year on each subsequent 1st January, until and including 1st January 2017. Calculate Joe's salary for each year from 2014 to 2017.
- Express the sequence of salaries from 2014 (term one) to 2017 (term four) using recursive notation.
37. \$1 500 is invested and accrues interest at a rate of 8% per annum, compounded annually. With this \$1500 as the first term in the sequence express the value of the account on this and each subsequent year as a sequence defined recursively.
38. A taxi charges a fixed amount of \$4.10 (called the flagfall) and then an additional \$1.69 per kilometre.
- Determine the fare for a taxi journey of (a) 10 km, (b) 20 km, (c) 30 km.
39. (Flat rate, or straight line, method of depreciation.)
- Over a period of 8 years a machine depreciates from its initial value of \$138 000, to its scrap value of \$26 000 at the end of the 8 years. If the amount depreciated each year stays the same determine the value of the machine after 1, 2 and 3 years.
40. (Unit cost, or units of production, method of depreciation.)
- A particular photocopier is expected to have a lifetime of 3 million copies, after which its initial value of \$27 000 will have reduced to a second hand value of just \$1 500. If this depreciation is at a fixed amount per copy made, find the value of the photocopier after 500 000, 1 000 000 and 1 500 000 copies.
- In its first year the photocopier makes 70 000 copies. What is its value at the end of this first year?
41. (Declining balance, or reducing balance, method of depreciation.)
- Each year the value of a car depreciates by 15% of its value at the beginning of that year. The car is initially worth \$36 000. With this \$36 000 as the first term express the value of the car on this and each subsequent year as a sequence defined recursively.

Jumping to later terms of arithmetic and geometric sequences.

Consider the arithmetic sequence defined by

$$T_{n+1} = T_n + 2 \text{ and } T_1 = 3$$

The rule allows us to obtain the terms of the sequence:

$$T_1 = 3$$

$$T_2 = T_1 + 2 = 3 + 2 = 5$$

$$T_3 = T_2 + 2 = 5 + 2 = 7$$

$$T_4 = T_3 + 2 = 7 + 2 = 9 \quad \text{etc.}$$

However, if we wanted to know the value of a term much later in the sequence, say T_{100} , it would be a tedious process to have to calculate all of the terms up to T_{100} . It would be more useful if we could jump to the desired term without having to determine all of the preceding ones.

Example 7

For the sequence defined recursively as $T_{n+1} = T_n + 7$ with $T_1 = 25$, determine the first four terms and the one hundredth term.

$$\begin{aligned} \text{With } T_{n+1} = T_n + 7 \quad \text{it follows that} \quad T_2 &= T_1 + 7 \\ &= 32. \\ T_3 &= T_2 + 7 \\ &= 39. \\ T_4 &= T_3 + 7 \\ &= 46. \end{aligned}$$

Note that for the second term we add 7 to T_1 *once*, for the third term we have added 7 to T_1 *twice*, for the fourth term we have added 7 to T_1 *three* times. It follows that for the one hundredth term we need to add 7 to T_1 ninety nine times.

$$\begin{aligned} \text{Hence } T_{100} &= T_1 + 99(7) \\ &= 25 + 693 \\ &= 718. \end{aligned}$$

The first four terms are 25, 32, 39 and 46 and the one hundredth term is 718.

If we apply the thinking used in the previous example to the general arithmetic sequence

$$a, \quad a + d, \quad a + 2d, \quad a + 3d, \quad a + 4d, \quad a + 5d, \quad \dots$$

we note that for T_2 the common difference, d , has been added once, for T_3 it has been added twice, for T_4 it has been added three times, and so on. Thus for T_n we need to add the common difference $(n - 1)$ times.

Thus the Arithmetic Progression

$$a, \quad a + d, \quad a + 2d, \quad a + 3d, \quad a + 4d, \quad a + 5d, \quad \dots$$

has an n^{th} term given by:

$$T_n = a + (n - 1) d$$

$$\text{Or } t_n = t_1 + (n - 1) d$$

Just pause for a moment and check that you understand the difference between a recursive formula, which tells you how each term is obtained from the previous term, and the formula for the n^{th} term, which allows you to jump to any term.

Note • Thinking of T_n as y , and n as x , the reader should see similarities between

$$T_n = a + (n - 1) d \quad \text{and} \quad y = mx + c.$$

This is no surprise given the linear nature of arithmetic sequences.

Example 8

For the AP: 11, 14, 17, 20, ...

Determine (a) T_{123} ,

(b) T_{500} ,

(c) which term of the sequence is the first to exceed 1 000 000.

$$\begin{aligned} \text{(a)} \quad T_{123} &= 11 + 122(3) \\ &= 377 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad T_{500} &= 11 + 499(3) \\ &= 1508 \end{aligned}$$

(c) Suppose that T_n is the first term to exceed 1 000 000.

$$\text{Now} \quad T_n = 11 + (n - 1) 3$$

$$\therefore 11 + (n - 1) 3 > 1\,000\,000$$

$$\text{i.e.} \quad n > 333\,330.\overline{6}$$

Thus the first term to exceed 1 000 000 is $T_{333\,331}$.

Example 9

An AP has a 50th term of 209 and a 61st term of 253. Find (a) the 62nd term,
(b) the 1st term.

(a) To go from T_{50} to T_{61} we must add the common difference 11 times.

$$\begin{array}{ll}
 \text{Thus if } d \text{ is the common difference then} & 253 - 209 = 11d \\
 \therefore & 44 = 11d \\
 \text{giving} & d = 4 \\
 \text{Hence} & T_{62} = T_{61} + 4 \\
 & = 257
 \end{array}$$

The 62nd term is 257.

(b) From our understanding of APs it follows that

$$\begin{array}{ll}
 T_{50} = T_1 + 49d & \\
 \therefore 209 = T_1 + 49(4) & \\
 209 - 196 = T_1 & \\
 T_1 = 13 &
 \end{array}$$

The 1st term is 13.

Alternatively we could use the given information to write $a + 49d = 209$
and $a + 60d = 253$
and then solve these equations simultaneously.

Example 10

For the sequence defined recursively as $T_{n+1} = 1.5T_n$ with $T_1 = 8192$,
determine the first four terms and the fifteenth term.

With $T_{n+1} = 1.5T_n$ it follows that

$$\begin{array}{ll}
 T_2 = 1.5T_1 & \\
 = 8192 \times 1.5 & \\
 = 12288 & \\
 T_3 = 1.5T_2 & \\
 = 12288 \times 1.5 & \\
 = 18432 & \\
 T_4 = 1.5T_3 & \\
 = 18432 \times 1.5 & \\
 = 27648 &
 \end{array}$$

Note that for the second term we multiply T_1 by 1.5 *once*, for the third term we have multiplied T_1 by 1.5 *twice*, for the fourth term we have multiplied T_1 by 1.5 *three* times. It follows that for the fifteenth term we need to multiply T_1 by 1.5 *fourteen* times.

$$\begin{array}{ll}
 \text{Hence } T_{15} = T_1 \times 1.5^{14} & \\
 = 8192 \times 1.5^{14} & \\
 = 2391484.5 &
 \end{array}$$

The first four terms are 8192, 12288, 18432 and 27648.
The fifteenth term is 2391484.5.



Make sure you can obtain this same value for the fifteenth term of the sequence of the previous example by using a calculator to display terms of the sequence.



Are the larger numbers displayed as shown on the right or does your calculator use scientific notation to display these numbers?



Can your calculator display the terms of the sequence graphically? Explore this possibility.



☒ $a_{n+1} = 1.5 \cdot a_n$ $a_1 = 8192$	
☐ $b_{n+1} = \square$ $b_1 = 0$	
☐ $c_{n+1} = \square$ $c_1 = 0$	
n	a_n
11	472392
12	708588
13	1062882
14	1594323
15	2391484.5

If we apply the thinking of the previous example to the general geometric sequence:

$a, \quad ar, \quad ar^2, \quad ar^3, \quad ar^4, \quad ar^5, \quad ar^6, \quad \dots$

we note that T_2 is ar^1 , T_3 is ar^2 , T_4 is ar^3 etc. Hence $T_n = ar^{n-1}$.

Thus the Geometric Progression

$a, \quad ar, \quad ar^2, \quad ar^3, \quad ar^4, \quad ar^5, \quad ar^6, \quad \dots$

has an n^{th} term given by:

$T_n = a \times r^{n-1}$	Or $t_n = t_1 r^{n-1}$
--------------------------	------------------------

Example 11

Determine the 12th term and the 15th term of the geometric sequence:

0.0025, 0.01, 0.04, 0.16, ...

By inspection the common ratio is 4.

Hence the 12th term will be 0.0025×4^{11}
 $= 10\,485.76$

and the 15th term will be 0.0025×4^{14}
 $= 671\,088.64$

Again make sure that you can obtain these same answers using the ability of some calculators to display the terms of a sequence.

Example 12

The 13th term of a GP is 12288 and the 16th term is 98304. Find (a) the 17th term,
(b) the 1st term.

- (a) To go from the 13th term to the 16th term we must multiply by the common ratio 3 times.

If r is the common ratio then

$$T_{16} = T_{13} \times r^3$$

$\therefore$

$$98304 = 12288 \times r^3$$

Giving

$$r = 2$$

Hence

$$\begin{aligned} T_{17} &= T_{16} \times 2 \\ &= 196608 \end{aligned}$$

The 17th term is 196608.

- (b) From our understanding of GPs it follows that $T_{13} = T_1 \times r^{12}$

i.e.

$$12288 = T_1 \times 2^{12}$$

Giving

$$T_1 = 3$$

The 1st term is 3.

(Alternatively we could write $ar^{12} = 12288$ and $ar^{15} = 98304$ and solve simultaneously.)

Growth and decay.

With their constant addition (or subtraction) arithmetic sequences describe linear growth (or decay).

With their repeated multiplication by r geometric sequences describe exponential growth, for $r > 1$, or decay, for $0 < r < 1$.

Consider the growth in the value of a house that is initially valued at \$500000 and is subject to an annual increase in value of 6.4%.

Initial value	= \$500000	←	T_1	
Value after 1 year	= \$500000 $\times 1.064$	←	T_2	
Value after 2 years	= \$500000 $\times 1.064^2$	←	T_3	
Value after 3 years	= \$500000 $\times 1.064^3$	←	T_4	etc.

These values form a geometric sequence with
and

$$\begin{aligned} T_{n+1} &= T_n \times 1.064 \\ T_1 &= 500000 \end{aligned}$$

Asked a question like: *At this rate how many years will it take for the value of this house to reach a value of \$1 000 000?*

we could display the terms of our sequence on a calculator or spreadsheet and view the progressive year by year values, as shown on the next page.

◆	A	B	C	D
1	Initial value			\$500,000.00
2	Percentage increase			6.40
3	Value at end of year		1	\$532,000.00
4			2	\$566,048.00
5			3	\$602,275.07
6			4	\$640,820.68
7			5	\$681,833.20
8			6	\$725,470.52
9			7	\$771,900.64
10			8	\$821,302.28
11			9	\$873,865.63
12			10	\$929,793.03
13			11	\$989,299.78
14			12	\$1,052,614.96

Create a spreadsheet like that shown above yourself.

The value of the house will be \$1 000 000 shortly after the end of the eleventh year, i.e. early in the 12th year.

However note carefully that in this situation, with the recursive definition

$$T_{n+1} = T_n \times 1.064 \quad \text{and} \quad T_1 = 500\,000$$

T_1 is the value after **zero** years. Hence we must remember that if we use the ability of a calculator to generate the terms of the sequence, according to the recursive rule given above, then the balance after n years will be given by T_{n+1} . I.e. in the calculator display above, $n = 13$ gives the value at the end of 12 years.

One way to avoid this possible source of confusion would be to use the ability of some calculators to accept a sequence defined using T_0 as the 1st term.

I.e. define the sequence as:

$$T_{n+1} = T_n \times 1.064 \quad \text{and} \quad T_0 = 500\,000,$$

as shown on the right.

Under such a definition T_n would indeed be the value after n years.

Alternatively we could use

$$T_{n+1} = T_n \times 1.064 \quad \text{and} \quad T_1 = 500\,000 \times 1.064$$

and again T_n would be the value after n years.

☒ $a_{n+1} = a_n \cdot 1.064$
 $a_1 = 500\,000$
☐ $b_{n+1} = \square$
 $b_1 = 0$
☐ $c_{n+1} = \square$
 $c_1 = 0$

n	a_n
9	8.2E+5
10	8.7E+5
11	9.3E+5
12	9.9E+5
13	1.1E+6

1052614.96443211

☒ $a_{n+1} = a_n \cdot 1.064$
 $a_0 = 500\,000$
☐ $b_{n+1} = \square$
 $b_0 = 0$
☐ $c_{n+1} = \square$
 $c_0 = 0$

n	a_n
8	8.2E+5
9	8.7E+5
10	9.3E+5
11	9.9E+5
12	1.1E+6

1052614.96443211

Exercise 4B

Without using the sequence display routine available on some calculators determine the one hundredth term in each of the following arithmetic sequences.

1. 11, 16, 21, 26, 31, 36,
2. -8, -5, -2, 1, 4, 7,
3. $T_{n+1} = T_n + 8$ with $T_1 = 23$.
4. $T_{n+1} = T_n - 2$ with $T_1 = 78$.

Without using the sequence display routine available on some calculators determine the twenty-fifth term in each of the following geometric sequences, leaving your answers in the form $a \times b^n$.

5. 5, 10, 20, 40, 80, 160,
6. 1.5, 6, 24, 96, 384, 1536,
7. $T_{n+1} = 3T_n$ with $T_1 = 8$.
8. $T_{n+1} = 2T_n$ with $T_1 = 11$.

Use the ability of some calculators to display the terms of a sequence to determine the requested term in each of the following sequences.

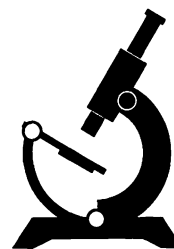
9. $T_{n+1} = T_n + 8$ with $T_1 = 7$. Determine T_{28} .
10. $T_{n+1} = 35 - 2T_n$ with $T_1 = 5$. Determine T_{20} .
11. $T_{n+1} = 3T_n + 2$ with $T_1 = 1$. Determine T_{19} .
12. $T_{n+1} = (-1)^n T_n + 3$ with $T_1 = 6$. Determine T_{45} .
13. Julie starts a new job at a factory manufacturing automobile components. The machine she operates requires several weeks before the operator is fully accustomed to it and so her output increases each day for the first 3 weeks (15 days). On the first day she successfully completes 48 items on the machine and increases this by 3 each day after that up to and including her 15th day on the machine.
Express the number of items completed on each of the first 15 days as a sequence using recursive notation.
How many items does she successfully complete on this 15th day on the machine?
14. Use the formula for the n^{th} term of an AP with common difference d and $T_1 = a$, i.e. the formula $T_n = a + (n - 1)d$, to explain why that for this AP, when we plot T_n on the y -axis and n on the x -axis the points obtained lie on a straight line of gradient d , and find the coordinates of the point where this straight line cuts the y -axis.
15. Use the formula for the n^{th} term of an GP with common ratio r and $T_1 = a$ to explain why that for this GP, when we plot T_n on the y -axis and n on the x -axis the points obtained fit an exponential curve (i.e. a curve with an equation of the form $y = hk^x$ for constant h and k) and find the equation of this curve and the coordinates of the point where it cuts the y -axis.

16. Write a few sentences explaining what happens to the terms of the following arithmetic progression as n gets very large, (i.e. as n "tends to infinity", which is written $n \rightarrow \infty$).
 $T_1 = a, \quad T_2 = a + d, \quad T_3 = a + 2d, \quad T_4 = a + 3d, \quad \dots \quad T_n = a + (n - 1)d, \quad \dots$
17. Write a few sentences explaining what happens to the terms of the following geometric progression as n gets very large, (i.e. as n "tends to infinity", which is written $n \rightarrow \infty$).
 $T_1 = a, \quad T_2 = ar, \quad T_3 = ar^2, \quad T_4 = ar^3, \quad \dots \quad T_n = ar^{n-1}, \quad \dots$
18. An arithmetic sequence has a first term of 8 and a common difference of 3. Determine the first four terms, the 50th term and the 100th term of the sequence.
19. An arithmetic sequence has a first term of 100 and a common difference of -3. Determine the first four terms, the 50th term and the 100th term of this sequence.
20. A geometric sequence has a first term of 11 and a common ratio of 2. Determine the first four terms, the 15th term and the 25th term of this sequence.
21. A geometric sequence has a first term of 2048 and a common ratio of 0.5. Determine the first four terms and the sixteenth term of this sequence.
22. Find an expression for T_n in terms of n for each of the following APs.
(a) 9, 15, 21, 27, 33, ... (b) 7, 8.5, 10, 11.5, 13, ...
23. Find an expression for T_n in terms of n for each of the following GPs.
(a) 3, 6, 12, 24, 48, ... (b) 100, 110, 121, 133.1, 146.41, ...
24. For the arithmetic sequence: 2, 9, 16, 23, ...
Determine (a) T_{123} , (b) T_{500} ,
(c) which term of the sequence is the first to exceed 1 000 000.
25. For the geometric sequence: 0.0026, 0.013, 0.065, 0.325, ...
Determine (a) T_{12} ,
(b) which term of the sequence is the first to exceed 1 000 000.
26. For the GP: 20 000 000, 15 000 000, 11 250 000, 8 437 500, ...
Determine (a) T_{12} , giving your answer to the nearest hundred,
(b) which term of the sequence is the first less than 1.
27. The n^{th} term of a sequence is given by $T_n = n^3$.
Obtain the first four terms of this sequence and state whether the sequence is arithmetic, geometric or neither of these.
28. An arithmetic sequence has a 19th term of 61 and a 41st term of 127.
Find (a) the 20th term, (b) the 1st term.

29. An arithmetic sequence has a 50th term of 1853 and a 70th term of 1793.
Find (a) the 51st term, (b) the 1st term.
30. A geometric sequence has a 10th term of 98 415 and a 13th term of 2 657 205.
Find (a) the 14th term, (b) the 1st term.
31. A geometric sequence has a 7th term of 28 672, a 9th term of 458 752 and a negative common ratio.
Find (a) the 10th term, (b) the 1st term.

The following questions involve situations in which a quantity is repeatedly multiplied by an amount (geometric progression) or has the same amount repeatedly added to it, or subtracted from it (arithmetic progression). However, do not be too quick to reach for the formulae for the n^{th} terms of such progressions. Instead think about what each situation requires you to find. This will often lead to the answer more quickly than simply "clutching" at formulae.

32. \$4000 is invested into an account paying interest at 8%, compounded annually. Determine (to the nearest cent) the amount in the account at the end of ten years.
33. (a) Scientists are studying a sample of bacteria and find that the initial population of 100 cells is doubling every hour. How many cells will there be in this sample after seven hours?
- (b) At the end of the 7 hours the scientists introduce a chemical to the sample which not only immediately stops the growth of new cells but also kills off cells at a rate of approximately 400 per hour, for each of the next 5 hours. How many cells are present in this sample at the end of these five hours?
- (c) Display the number of cells over this twelve hour period as a graph.
34. (a) The value of an item of machinery is expected to depreciate at a constant \$5000 every year. Assuming this to be the case what will be the value of this item of machinery after 10 years if it is initially worth \$98000 ?
- (b) By the end of each year the value of an item of machinery is expected to depreciate by 5% of its value at the beginning of that year. Assuming this to be the case what will be the value of this item of machinery after 10 years, to the nearest \$1000, if at the start of this period it is worth \$98 000 ?



35. The population of a country is growing exponentially. The populations in 2010, 2011, 2012 and 2013 were as follows.

Year	2010	2011	2012	2013
Population	45 million	45.8 million	46.6 million	47.5 million

Show that these figures support the theory that the population is growing at an annual rate of approximately 1.8%.

Predict the population for this country for the year 2027.

36. The population of a particular species of animal is declining exponentially. The numbers of these animals thought to be in existence in the wild in 2010, 2011, 2012 and 2013 were as follows:

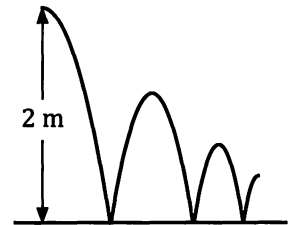
Year	2010	2011	2012	2013
Number	18 000	16 500	15 200	14 000

Show that these figures support the theory that the number of these animals thought to be in existence is declining at an annual rate of approximately 8%.

Predict how many of these animals will exist in the wild in the year 2023.

37. When a particular ball is dropped onto a horizontal surface the height it reaches on its first bounce is 60% of the height from which it was dropped. Subsequent bounce heights are each 60% of the height of the previous bounce. If the ball is dropped from a height of 2 metres, onto a horizontal surface, find

- the height reached on the first bounce,
- the height reached on the sixth bounce.



38. An investor purchases a valuable painting for \$2 500 000. Experts predict that the value of the painting will rise exponentially at a rate of 4.6% per annum (i.e. the experts predict that by the end of each year the value of the painting will have risen by 4.6% of its value at the beginning of that year).
- Predict, to the nearest \$10 000, the value of this painting after 5 years.
 - Predict, to the nearest \$10 000, the value of this painting after 10 years.
 - Approximately how many years will it take for the value of the painting to reach \$5 000 000 ?
39. If a house currently valued at \$600 000 were to gain in value at 5.6% per annum, compounded annually, when would its value first exceed two million dollars?

First order linear recurrence relations.

Arithmetic sequences have recurrence rules of the form $T_1 = a, T_{n+1} = T_n + d$.

Geometric sequences have recurrence rules of the form $T_1 = a, T_{n+1} = r \times T_n$.

We will now consider recurrence rules of the form $T_1 = a, T_{n+1} = b \times T_n + c$.

This last form is an example of a *first-order linear recurrence relation*.

- Arithmetic sequences and geometric sequences are also of the third form given above. With $b = 1$ and $c = d$ we have the AP rule and with $b = r$ and $c = 0$ we have the GP rule. However we will now consider more general examples of the third form.
- If the relationship involved the two previous terms, eg $T_{n+2} = b \times T_{n+1} + c \times T_n + d$ then it would be a *second-order linear recurrence relation*. We will concentrate our attention on first order linear recurrence relations.

Consider recurrence relations for which each term, after the first, is obtained from the previous term by multiplying by some constant value, b , and then adding a constant value, c .

For example $T_1 = 400, T_{n+1} = 1.2 \times T_n - 50$.

In this case $T_1 = 400$

$$T_2 = 1.2 \times T_1 - 50$$

$$= 1.2 \times 400 - 50$$

$$= 430$$

$$T_3 = 1.2 \times T_2 - 50$$

$$= 1.2 \times 430 - 50$$

$$= 466$$

$$T_4 = 1.2 \times T_3 - 100$$

$$= 1.2 \times 466 - 50$$

$$= 509.2$$

You might be wondering when such a first-order recurrence relationship would occur in real life.

Well let us consider the situation of a breeder of animals who, as well as increasing the size of his herd of these animals, also wants to sell some of his stock each year to other breeders.

Suppose the breeder has a herd of 400 animals and that through his successful breeding program this number is increasing at a rate of 20% per year. Suppose also that the breeder plans to sell 50 of the animals at the end of each year.

Initially his herd numbers 400
One year later it will number: $400 \times 1.2 - 50$
 $= 430$

After another year it will be $430 \times 1.2 - 50$
 $= 466$

I.e. the recurrence relation considered previously (although the need to round to whole numbers would see the real situation deviate a little from the theoretical model).

☒ $a_{n+1} = 1.2 \cdot a_n - 50$ $a_1 = 400$													
☐ $b_{n+1} = \square$ $b_1 = 0$													
☐ $c_{n+1} = \square$ $c_1 = 0$													
<table style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="text-align: center; border-bottom: 1px solid black;">n</th> <th style="text-align: center; border-bottom: 1px solid black;">a_n</th> </tr> </thead> <tbody> <tr> <td style="text-align: center;">1</td> <td style="text-align: center;">400</td> </tr> <tr> <td style="text-align: center;">2</td> <td style="text-align: center;">430</td> </tr> <tr> <td style="text-align: center;">3</td> <td style="text-align: center;">466</td> </tr> <tr> <td style="text-align: center;">4</td> <td style="text-align: center; border: 1px solid black;">509.2</td> </tr> <tr> <td colspan="2" style="border: 1px solid black; padding: 2px;">509.2</td> </tr> </tbody> </table>		n	a _n	1	400	2	430	3	466	4	509.2	509.2	
n	a _n												
1	400												
2	430												
3	466												
4	509.2												
509.2													

Exercise 4C.

1. Determine the first four terms for the sequence defined recursively as follows:

$$T_1 = 10, \quad T_{n+1} = 2 \times T_n + 3.$$

2. Determine the first four terms for the sequence defined recursively as follows:

$$T_1 = 200, \quad T_{n+1} = 3 \times T_n - 150.$$

3. Determine the first four terms for the sequence defined recursively as follows:

$$T_1 = 200, \quad T_{n+1} = 2 \times T_n - 250.$$

4. For each of the sequences A, B and C defined below:

Sequence A: $T_1 = 5, \quad T_{n+1} = 2 \times T_n + 1.$

Sequence B: $T_1 = 30, \quad T_{n+1} = 2 \times T_n - 30.$

Sequence C: $T_1 = 90, \quad T_{n+1} = 2 \times T_n - 95.$

(a) Determine the first five terms, T_1 to T_5 .

(b) With T_n on the vertical axis and n on the horizontal axis, display T_1 to T_5 graphically.

(c) Choose the most appropriate label from the following list:

Increasing sequence. Decreasing sequence. Steady state sequence.

5. Use your calculator to view the first 100 terms of the following recursively defined sequence both numerically and graphically.

$$T_1 = 35000, \quad T_{n+1} = 0.96 \times T_n + 2000.$$

What happens to the terms of the sequence "in the long term", i.e. as $n \rightarrow \infty$?

Explain why this "long term steady state situation" occurs.

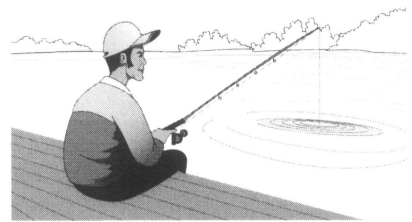
6. A lakes complex is thinking of allowing some very limited fishing to occur in order to keep the population of fish in the lakes under control.

It is thought that the current population of fish is approximately 2000 and that if left uncontrolled this will increase at approximately 12% per year.

If limited fishing is allowed one suggestion is to

allow approximately 300 fish to be caught near the end of each 12 month period with the first "catch" occurring 12 months from now.

According to these figures, and if the limited fishing were allowed as suggested, what would the population of fish be just after the "catch" in 1 year, 2 years, 5 years and 10 years?



7. A country has
- a current population of 17.6 million,
 - a population growth rate (i.e. the birth rate take away the death rate) of approximately 1.2%
- and
- takes in roughly 120 000 people per year under its immigration program.

Writing the population of this country, n years from now, as T_n , write a recursive expression for T_n in terms of T_{n-1} , and state T_0 .

Determine estimates for T_5 and T_{10} .

Miscellaneous Exercise Four.

This miscellaneous exercise may include questions involving the work of this chapter, the work of any previous chapters, and the ideas mentioned in the preliminary work section at the beginning of the book.

- Write the next two entries in each of the following sequences
 - 3, 9, 27, 81,,
 - A, E, I,,
 - M, T, W, T, F,,
 - 32, 16, 8, 4, 2, 1,,
- Find the first six terms (T_1 to T_6) for each of the following.
 - $T_{n+1} = 2T_n + 3$, $T_1 = 7$
 - $T_{n+1} = 2T_n - 1$, $T_2 = 11$
 - $T_{n+1} = 0.5T_n$, $T_3 = 64$
 - $T_{n+1} = T_n + 7$, $T_4 = 26$
- For the sequence 1, 3, 11, 43, 171, 683, 2731, 10923, ...find
 - T_4 ,
 - $2T_3$,
 - $3T_1 + T_2$,
 - $T_3 - T_2$.
 - The sequence has a recursive formula of the form $T_n = aT_{n-1} + b$.
Find a and b and hence determine T_9 and T_{10} .
- An arithmetic sequence has a first term of 10 and a common difference of 3.
Determine the 50th term of this sequence.
- A geometric sequence has a first term of 1536 and a common ratio of 0.5.
Determine the 14th term of this sequence.
- A study of a number of adults found that their heights, h cm, and their weights, w kg, could be reasonably well modelled by the equation

$$w = 0.9h - 80$$
 Interpret the number 0.9 for this group of people.
- Determine r_{xy} , the coefficient of linear correlation, and $y = A + Bx$, the equation of the linear regression line, for the following data.

x	49	26	29	40	34	22	35	37	40	29
y	54	41	44	50	44	42	51	48	55	47

 Hence predict the value of y for (a) $x = 32$, (b) $x = 60$,
and comment on the validity of your predicted values.
- With the aid of a calculator able to display the terms of recursively defined sequences determine what happens to the terms of the following sequences "in the long term", i.e. as $n \rightarrow \infty$?
 - $T_1 = 5000$, $T_{n+1} = 0.8 \times T_n + 800$.
 - $T_1 = 5000$, $T_{n+1} = 0.8 \times T_n + 1600$.
 - $T_1 = 5000$, $T_{n+1} = 1.2 \times T_n - 800$.

9. \$4000 is invested into an account paying interest at 8%, compounded annually and \$200 is withdrawn from the account after each 12 months. Thus:

$$\text{Amount in account at end of 1 yr} = \$4000 \times 1.08 - \$200 \quad \leftarrow T_1$$

$$\text{Amount in account at end of 2 yrs} = (\$4000 \times 1.08 - \$200) \times 1.08 - \$200 \quad \leftarrow T_2$$

Express T_{n+1} in terms of T_n and, with the assistance of a calculator able to display the terms of a recursively defined sequence, determine (nearest cent) the amount in the account at the end of ten years after the \$200 for the year has been withdrawn.

10. The table below gives the number of motor vehicles and the number of televisions owned per 1000 people in eighteen countries.

Country	A	B	C	D	E	F	G	H	I
N ^o . of motor vehicles (V)	714	450	203	361	570	92	748	202	528
N ^o . of televisions (T)	731	618	412	542	486	215	815	271	560

Country	J	K	L	M	N	O	P	Q	R
N ^o . of motor vehicles (V)	613	72	462	46	173	416	73	215	155
N ^o . of televisions (T)	641	280	431	168	225	432	201	402	248

Display the above information on a scattergraph and use your graph to predict the number of televisions per 1000 people for a nineteenth country given that this nineteenth country has 285 motor vehicles per 1000 people.

Comment on the likely reliability of your prediction.

What percentage of the variation in the number of televisions can be explained by the variation in the number of motor vehicles?

11. A survey was conducted to investigate whether the frequency with which an adult engaged in sport or exercise was associated with gender. A number of people were asked how many times they engaged in sport or exercise in the course of an average week. The responses gave rise to the following table:

	Never	Once per week	Twice per week	Three times per week	Four times per week	Five times per week
Male	36	15	18	27	30	24
Female	52	18	31	33	46	40

Re draw the table showing either column percentages or row percentages as appropriate and determine, with justification, whether or not the results suggest that the frequency with which an adult engages in sport or exercise is associated with gender.